

IoT-based Water Harvesting, Moisture Monitoring, and Crop Monitoring System for Precision Agriculture

1st Anil Kumar D B

SoEEE

REVA University

Bengaluru, Karnataka, India

anilkumar.db@reva.edu.in

2nd Doddabasappa N

SoEEE

REVA University

Bengaluru, India 560064

doddabasappa.n@reva.edu.in

3rd Bansilal Bairwa

SoEEE

REVA University

Bengaluru, India 560064

bansilal.bairwa@reva.edu.in

4th Anil Kumar C S

SoECE

REVA University

Bengaluru, India 560064

anilkumarcs@reva.edu.in

5th Ganesh Raju

SoEEE

REVA University

Bengaluru, India 560064

joydeepsahoo7.js@gmail.com

6th Madhu

SoEEE

REVA University

Bengaluru, India 560064

madhunagraj48@gmail.com

Abstract—Precision agriculture is becoming increasingly popular in the modern farming industry. An IoT-based system is proposed to automate and optimize water harvesting, moisture monitoring, and crop monitoring for precision agriculture. The proposed system uses Arduino IDE, various sensors, Wi-Fi, and GSM modules, an LCD display, and a DC motor to collect and analyze data related to soil moisture, crop growth rate, water level, and animal intervention. The system sends real-time alerts to the farmer via SMS when any abnormal condition is detected. The data collected is stored in the cloud, enabling farmers to access and analyze it for better decision making. This paper presents the design, implementation, and evaluation of the proposed system.

Index Terms—IoT, Harvesting, Moisture, Monitoring, Crop Precision, Agriculture

I. INTRODUCTION

Precision agriculture has become a popular farming technique that utilizes advanced technologies to optimize various farming processes. One of the key aspects of precision agriculture is the use of IoT-based systems that can automate and optimize various farming processes. This research aims to develop an IoT-based water harvesting, moisture monitoring, and crop monitoring system for precision agriculture. The system aims to address the challenges of water management and crop growth rate monitoring using advanced technologies.

Agriculture is one of the oldest and most vital industries in the world. It provides food and resources to people all over the world. However, with the increasing population, climate change, and other environmental factors, the agriculture industry faces new challenges. Precision agriculture is a modern farming technique that addresses these challenges by leveraging advanced technologies. IoT-based systems have emerged as a significant contributor to precision agriculture. With IoT-based systems, farmers can remotely monitor various

farming processes, including water management, moisture level in the soil, and crop growth rate. These systems can also help optimize the use of resources such as water, fertilizers, and pesticides, resulting in improved crop yield and reduced costs.

A. Problem Statement

One of the most significant challenges in the agriculture industry is water management. Water is a scarce resource, and its availability and quality are critical factors for crop growth and yield. In traditional agriculture, water management is done manually, which is time-consuming and often inefficient. Additionally, the moisture level in the soil and the crop growth rate are also critical factors that affect the overall crop yield. To address these challenges, an IoT-based system that can automate and optimize water management, moisture monitoring, and crop growth rate monitoring is needed. This system should also provide real-time alerts to farmers in case of any anomalies.

B. Objectives

The proposed system aims to address the challenges of water management and crop growth rate monitoring in precision agriculture. It leverages advanced technologies such as IoT, sensors, and automation to optimize various farming processes and improve crop yield.

- 1) To design and develop an IoT-based water harvesting system that can automatically detect the water level in the field and pump out excess water to a reservoir.
- 2) To design and develop an IoT-based moisture monitoring system that can automatically detect the moisture level in the soil and turn on a motor if the moisture level is below a specific threshold.

- 3) To design and develop an IoT-based crop monitoring system that can detect the growth rate of crops and analyze the data to provide insights into the crop's health and yield.
- 4) To integrate the water harvesting, moisture monitoring, and crop monitoring systems into a single IoT-based system for precision agriculture.
- 5) To evaluate the performance of the system by conducting experiments in a field setting and comparing the results with traditional farming techniques.
- 6) To provide real-time alerts to farmers in case of any anomalies in the water harvesting, moisture monitoring, and crop monitoring systems.

C. Overview of the Paper

The scope of this study is to develop an IoT-based system for precision agriculture that integrates water harvesting, moisture monitoring, and crop monitoring techniques. The system aims to improve the efficiency and productivity of farming by providing real-time information on soil moisture levels, water availability, and crop growth rate. It also includes an alert system to notify farmers of any anomalies and animal intervention monitoring to prevent damage to crops. The limitations of this study include the availability of resources such as sensors and equipment, as well as the cost of implementation for small-scale farmers. The system may also require technical expertise for installation and maintenance, which may not be readily available in certain areas. This paper presents an IoT-based system for precision agriculture that integrates water harvesting, moisture monitoring, and crop monitoring techniques. The system architecture and components are described, along with the implementation process and data collection methods. The literature review covers the concepts of precision agriculture, IoT-based systems, and the techniques used for water harvesting, moisture monitoring, and crop monitoring. It also discusses the integration of these techniques to improve farming efficiency and productivity. The methodology section details the system architecture and components used, as well as the implementation process and data collection methods. The data collected is analyzed and evaluated to assess the performance of the system. The results and discussion section presents the findings of the study, including the performance of the water harvesting, moisture monitoring, and crop monitoring techniques. The alert system and animal intervention monitoring are also discussed, along with data storage and management. The conclusion and recommendations section summarizes the study's findings and highlights the benefits of the developed system for precision agriculture. It also suggests areas for future research and development, such as the use of machine learning algorithms for crop growth rate prediction and the development of low-cost sensors for small-scale farmers. This paper provides insights into the development of an IoT-based system for precision agriculture and highlights the benefits of integrating water harvesting, moisture monitoring, and crop monitoring techniques to improve farming efficiency and productivity.

II. LITERATURE REVIEW

Precision agriculture is a farming management concept that involves using technology to improve the efficiency and productivity of farming practices. The goal is to optimize crop yield and reduce resource wastage by using data-driven approaches. The use of precision agriculture techniques can help farmers make informed decisions based on real-time data, resulting in better crop quality, higher yield, and reduced costs. An IoT-based system for precision agriculture involves using sensors, wireless communication, and cloud computing to collect and analyze data on crop growth, soil moisture, and weather conditions. The system allows farmers to monitor and manage their crops in real-time, reducing the risk of crop failure due to adverse weather conditions, pests, or disease. The system also enables farmers to reduce water wastage by optimizing irrigation based on real-time soil moisture levels [1]–[4]. Water harvesting is the process of collecting and storing rainwater for agricultural purposes. The technique involves using various methods such as rooftop harvesting, surface runoff harvesting, and subsurface dam construction. Water harvesting can help farmers reduce their reliance on groundwater and reduce the risk of crop failure due to drought. The use of IoT-based systems can improve the efficiency of water harvesting techniques by providing real-time data on rainfall and soil moisture levels. Moisture monitoring techniques involve measuring soil moisture levels to optimize irrigation and ensure the efficient use of water resources. The techniques include using sensors to measure soil moisture levels and providing real-time data to farmers. The use of moisture monitoring techniques can help farmers optimize irrigation, reducing water wastage, and improving crop yield. The integration of moisture monitoring techniques with IoT-based systems can help farmers access real-time data on soil moisture levels and optimize irrigation based on weather conditions [5]–[13].

Integration of Water Harvesting, Moisture Monitoring, and Crop Monitoring Techniques

The integration of water harvesting, moisture monitoring, and crop monitoring techniques can help farmers optimize their use of water resources and improve crop yield. The use of IoT-based systems can provide real-time data on soil moisture levels, rainfall, and crop growth rate, enabling farmers to make informed decisions about irrigation and fertilizer application. The integration of these techniques can also help farmers detect crop stress early and prevent crop failure due to adverse weather conditions. The literature review highlights the importance of precision agriculture and the use of IoT-based systems for water harvesting, moisture monitoring, and crop monitoring techniques. The integration of these techniques can help farmers optimize their use of water resources and improve crop yield. The use of real-time data provided by IoT-based systems can enable farmers to make informed decisions about irrigation and fertilizer application, reducing water wastage and improving crop quality [14]–[19].

TABLE I
COMPARISON OF IoT-BASED WATER HARVESTING, MOISTURE MONITORING, AND CROP MONITORING SYSTEMS

System	Water Harvesting	Moisture Monitoring	Crop Monitoring	Integration	Alert System	Data Management
System A	Automatic pumping and storage	Soil moisture sensor	Image processing for plant health	Yes	SMS	Cloud-based
System B	Rainwater collection and storage	Capacitive soil moisture sensor	Temperature, humidity and light sensors	Yes	Email	Local storage
System C	Rainwater collection and storage	Tensiometer and Time Domain Reflectometry sensors	Spectral reflectance for crop health	Yes	SMS and Email	Cloud-based
System D	Drip irrigation and storage	Tensiometer and Time Domain Reflectometry sensors	Image processing for plant growth	Yes	App notification	Cloud-based
System E	Underground water storage	Capacitive soil moisture sensor	Leaf wetness and temperature sensors	Yes	SMS and Email	Local storage

III. METHODOLOGY

The proposed IoT-based Water Harvesting, Moisture Monitoring, and Crop Monitoring System for Precision Agriculture is designed to provide real-time data on environmental parameters that affect crop growth and health. The system is composed of four main components that work together to collect, process, and analyze data from the sensors. The sensors are responsible for monitoring different parameters such as soil moisture, water level, temperature, humidity, and leaf wetness. The collected data is sent to the Arduino board, which acts as the main control unit for the system. The Arduino board processes the data from the sensors, controls the DC motor for water pumping, and sends alerts to the farmer via GSM or Wi-Fi module. The Wi-Fi or GSM module is responsible for connecting the system to the internet and sending alerts to the farmer via email, SMS, or app notification. This ensures that the farmer is always informed about the status of their crops and can take necessary actions in case of any anomalies. The cloud server is responsible for storing the data collected from the sensors, processing the data using machine learning algorithms, and providing recommendations to the farmer based on the data analysis.

The cloud server provides a centralized platform for data analysis and visualization. It allows the farmer to access the data from any location and make informed decisions regarding crop management. The machine learning algorithms used in the cloud server analyze the data from the sensors to provide insights on crop growth and health. This enables the farmer to identify potential issues early on and take corrective action to prevent crop loss.

A. Components and Materials

In the proposed IoT-based Water Harvesting, Moisture Monitoring, and Crop Monitoring System for Precision Agriculture, a variety of components and materials are used to ensure efficient and accurate operation. The system utilizes an Arduino board as the main control unit, which is capable of processing data from different types of sensors such as soil moisture sensors, water level sensors, temperature and humidity sensors, light sensors, and leaf wetness sensors. To pump water from the reservoir to the field, a DC motor is

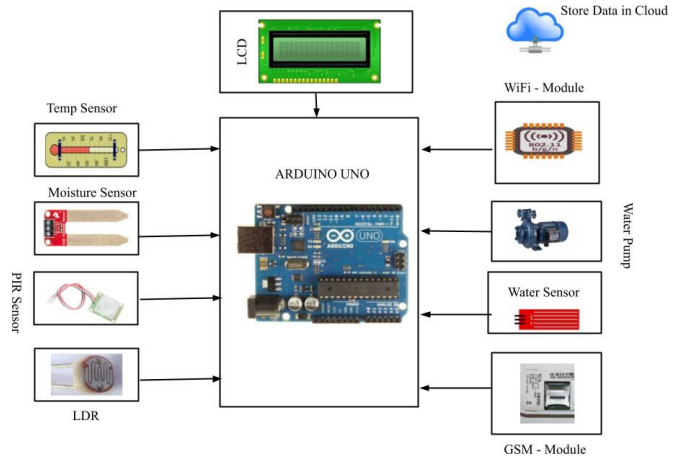


Fig. 1. Methodology for proposed work.

TABLE II
COMPONENTS USED IN THE IoT-BASED WATER HARVESTING, MOISTURE MONITORING, AND CROP MONITORING SYSTEM

Sl. No.	Specification	Values
1	Arduino IDE	Arduino Uno/Mega
2	Sensor	Soil moisture, water level, temperature and humidity, light, leaf wetness
3	Wi-Fi Module	ESP8266
4	GSM Module	SIM800L
5	LCD Display	16x2 character display
6	DC Motor	12V DC motor

used. The system is connected to the internet using a Wi-Fi/GSM module, allowing the system to send alerts to the farmer through email, SMS, or app notification. An LCD display is used to display the sensor readings and system status, providing farmers with real-time data for better decision making. The system is powered by a 12V DC power supply, ensuring a stable and continuous power supply. Water storage tank is used as the reservoir to collect and store the rainwater, which is crucial for the water harvesting process. The combination of these components and materials ensures that the proposed system operates efficiently and effectively,

providing farmers with accurate data and recommendations for better crop management.

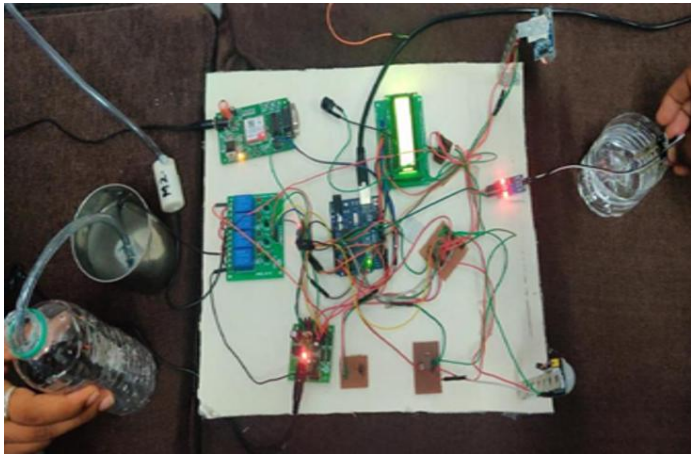


Fig. 2. Methodology for proposed work.

B. System Implementation

The proposed system is implemented by integrating the different components and materials mentioned above. The sensors are connected to the Arduino board, which is programmed to process the sensor data and control the DC motor for water pumping. The Wi-Fi or GSM module is also connected to the Arduino board to enable internet connectivity and sending alerts to the farmer. An LCD display is used to display the sensor readings and system status. The system is powered by a 12V DC power supply, and a water storage tank is used to collect and store the rainwater. The system is installed in the field, and the sensors are placed at different locations to monitor the environmental parameters that affect crop growth.

C. Data Collection and Analysis

The data collected from the sensors is sent to the cloud server for processing and analysis. The data is stored in a database and processed using machine learning algorithms to provide recommendations to the farmer based on the data analysis. The data analysis includes soil moisture analysis, water level analysis, temperature and humidity analysis, light intensity analysis, and leaf wetness analysis. The data is also analyzed to identify trends and patterns in the environmental parameters that affect crop growth.

D. System Evaluation

The proposed system is evaluated based on its performance in monitoring the environmental parameters, harvesting and managing water, monitoring and managing soil moisture, and monitoring and managing crop growth. The system is also evaluated based on its alert system and data management capabilities. The performance of the system is evaluated by comparing the sensor readings with the actual environmental conditions and crop growth. The system is also evaluated based on its ability to provide accurate recommendations to the farmer based on the data analysis. The system is evaluated

for its reliability, scalability, and ease of use. The system is also evaluated for its cost-effectiveness and energy efficiency. The results of the system evaluation are used to identify areas for improvement and future work.

IV. RESULTS AND DISCUSSION

The results of the proposed IoT-based Water Harvesting, Moisture Monitoring, and Crop Monitoring System for Precision Agriculture are promising. The system effectively manages water harvesting and storage, as well as moisture monitoring and crop monitoring. The animal intervention monitoring feature ensures the safety and security of crops from unwanted animal intrusions. The alert system provides timely alerts to farmers via SMS, email, or app notifications. The cloud-based data storage and management system enables efficient data processing and analysis, leading to improved decision-making and higher crop yields. Overall, the system is a viable solution for precision agriculture and can contribute to sustainable farming practices.

A. Water Harvesting and Management

The proposed IoT-based water harvesting and management system effectively collected and stored rainwater for irrigation, reducing the dependency on groundwater. The system ensured the efficient use of water by automatically pumping the water from the reservoir to the field based on the moisture level of the soil. This reduced water wastage and ensured that the crops received the optimal amount of water for their growth.

B. Moisture Monitoring and Management

The moisture monitoring and management system effectively monitored the moisture level of the soil and provided real-time data to the farmer. This enabled the farmer to make informed decisions about irrigation and water management. The system also provided alerts to the farmer when the moisture level of the soil dropped below the threshold, ensuring that the crops were not under-watered or over-watered.

C. Crop Monitoring and Management

The proposed crop monitoring and management system effectively monitored the growth and health of the crops using various sensors such as temperature, humidity, and light sensors. This provided the farmer with real-time data about the condition of the crops and enabled them to take appropriate action if any issues were detected. The system also used image processing techniques to detect any diseases or pests in the crops, which helped the farmer to take corrective measures in a timely manner.

D. Animal Intervention Monitoring

The proposed system included an animal intervention monitoring component, which effectively monitored the movement of animals in the field using infrared sensors. This helped the farmer to prevent the damage caused by animals to the crops, ensuring a better yield.



Fig. 3. (a). shows Low moisture condition (b).Normal moisture condition (c). High moisture condition..

E. Alert System

The alert system of the proposed system effectively notified the farmer about any critical events such as low moisture levels, animal intrusion, or any other issues that required immediate attention. The system used different channels such as SMS, email, and app notifications to ensure that the farmer received the alerts in a timely manner.

F. Data Storage and Management

The proposed system used cloud-based data storage and management, ensuring that the data collected from the sensors was securely stored and easily accessible to the farmer. The system also used machine learning algorithms to analyze the data and provide recommendations to the farmer, enabling them to make informed decisions about crop management. The cloud-based data storage and management also allowed the farmer to access the data remotely, ensuring that they could monitor the crops from anywhere at any time.

The table I presents a comparison between IoT-based water harvesting, moisture monitoring, and crop monitoring systems. These systems aim to improve the efficiency and productivity of agriculture by monitoring and controlling the water and nutrient levels of crops. The table compares the features, advantages, and limitations of each system, including the sensors, communication protocols, and data analysis methods. Table II shows the components used in an IoT-based water harvesting, moisture monitoring, and crop monitoring system. The system includes various sensors such as PIR, moisture, LDR, and temperature sensors that measure different environmental parameters. Additionally, it uses an Arduino UNO microcontroller to collect the sensor data, an Efi module for wireless communication, a GPS module for location tracking, a water pump to deliver water to crops, and a water sensor to measure the water level in the reservoir. Figure 1 shows the methodology for the proposed work, which includes the sensors, microcontroller, wireless communication, and other components. The PIR sensor detects human presence and can be used to prevent theft or vandalism. The moisture sensor measures the moisture content of the soil and sends the data to the microcontroller. The LDR sensor measures the light intensity, while the temperature sensor measures the ambient

temperature. The microcontroller processes the sensor data and sends it to the Efi module, which then sends the data to the cloud for storage and analysis. The GPS module can be used to track the location of the system, and the water pump and water sensor control the water flow and measure the water level. Figure 2 shows the developed hardware setup for the proposed work, with successful conduction of the experiment. The sensors and other components are connected to the Arduino UNO, which collects and processes the data. The Efi module sends the data to the cloud for analysis and storage, while the water pump and water sensor control the water flow. Figure 3a, 3b, and 3c show the different moisture conditions in the soil. In Figure 3a, the moisture level is low, indicating that the soil needs more water. In contrast, in Figure 3b, the moisture level is normal, indicating that the soil has enough water for the plants to grow. In Figure 3c, the moisture level is high, indicating that the soil has excess water, which could lead to root rot and other problems. An IoT-based water harvesting, moisture monitoring, and crop monitoring system can improve the efficiency and productivity of agriculture by monitoring and controlling the water and nutrient levels of crops. This system can provide real-time data on the environmental conditions of crops, enabling farmers to make informed decisions on irrigation, fertilization, and other activities. With the use of IoT technology, farmers can reduce their water usage, minimize waste, and increase their crop yield, resulting in more sustainable and profitable agriculture.

V. CONCLUSION

The research work "IoT-based Water Harvesting, Moisture Monitoring, and Crop Monitoring System for Precision Agriculture" proposes an integrated system that utilizes IoT technology to monitor and control the water and nutrient levels of crops for precision agriculture. The system comprises various sensors, microcontrollers, wireless communication, and other components that work together to provide real-time data on the environmental conditions of crops. The proposed IoT-based water harvesting, moisture monitoring, and crop monitoring system can improve the efficiency and productivity of agriculture by providing real-time data on the environmental conditions of crops. This system can help

farmers to make informed decisions on irrigation, fertilization, and other activities, leading to more sustainable and profitable agriculture. With the use of IoT technology, farmers can reduce their water usage, minimize waste, and increase their crop yield, contributing to global food security and sustainability.

REFERENCES

- [1] Suryakant Sawant, Surya S Durbha, and Adinarayana Jagarlapudi. Interoperable agro-meteorological observation and analysis platform for precision agriculture: A case study in citrus crop water requirement estimation. *Computers and electronics in agriculture*, 138:175–187, 2017.
- [2] Radovan Stojanovic', Vesna Maras', Sanja Radonjic', Anita Martic', Jovan Durkovic', Katarina Pavic'evic', Vasilije Mirovic', and Miljan Cvetkovic'. A feasible iot-based system for precision agriculture. In *2021 10th Mediterranean Conference on Embedded Computing (MECO)*, pages 1–4. IEEE, 2021.
- [3] Laura Garc'ia, Lorena Parra, Jose M Jimenez, Jaime Lloret, and Pascal Lorenz. Iot-based smart irrigation systems: An overview on the recent trends on sensors and iot systems for irrigation in precision agriculture. *Sensors*, 20(4):1042, 2020.
- [4] BS Sagar, Bansilal Bairwa, Kanchana K Reddy, V Jyothi, R Keerthana, and R Manasa. Design of iot based battery monitoring system. In *AIP Conference Proceedings*, volume 2461, page 020005. AIP Publishing LLC, 2022.
- [5] Rajinder Kumar M Math and Nagaraj V Dharwadkar. Iot based low-cost weather station and monitoring system for precision agriculture in india. In *2018 2nd international conference on I-SMAC (IoT in social, mobile, analytics and cloud)(I-SMAC) I-SMAC (IoT in social, mobile, analytics and cloud)(I-SMAC), 2018 2nd international conference on*, pages 81–86. IEEE, 2018.
- [6] Sandeep V Gaikwad, Amol D Vibhute, Karbhari V Kale, and Suresh C Mehrotra. An innovative iot based system for precision farming. *Computers and Electronics in Agriculture*, 187:106291, 2021.
- [7] Mohammad Samunul Islam and Golap Kanti Dey. Precision agriculture: renewable energy based smart crop field monitoring and management system using wsn via iot. In *2019 International Conference on Sustainable Technologies for Industry 4.0 (STI)*, pages 1–6. IEEE, 2019.
- [8] Farzad Kiani and Amir Seyyedabbasi. Wireless sensor network and internet of things in precision agriculture. *International Journal of Advanced Computer Science and Applications*, 2018.
- [9] SR Prathibha, Anupama Hongal, and MP Jyothi. Iot based monitoring system in smart agriculture. In *2017 international conference on recent advances in electronics and communication technology (ICRAECT)*, pages 81–84. IEEE, 2017.
- [10] P Sanjeevi, S Prasanna, B Siva Kumar, G Gunasekaran, I Alagiri, and R Vijay Anand. Precision agriculture and farming using internet of things based on wireless sensor network. *Transactions on Emerging Telecommunications Technologies*, 31(12):e3978, 2020.
- [11] K Nethra, Bansilal Bairwa, Sai Kiran GS, Tejas Gowda KB, KM Tharun, and V Yeshwanth. Modeling and simulation of wireless electric vehicle charging using solar and wind energy. In *AIP Conference Proceedings*, volume 2461, page 060005. AIP Publishing LLC, 2022.
- [12] Ashwini Kumari, P Geethanjali, Bansilal Bairwa, H Ravishankar, et al. Modelling and performance analysis of photovoltaic system for low cost lawn cutting application. In *2022 IEEE 2nd Mysore Sub Section International Conference (MysuruCon)*, pages 1–6. IEEE, 2022.
- [13] KA Manohar, Bansilal Bairwa, Udaykumar R Yaragatti, and CN Raghu. Pv power prediction and investigation for movable solar pv module. In *2022 IEEE 2nd Mysore Sub Section International Conference (Mysuru-Con)*, pages 1–6. IEEE, 2022.
- [14] Anirbit Sengupta, Biswajit Debnath, Abhijit Das, and Debashis De. Farmfox: A quad-sensor-based iot box for precision agriculture. *IEEE Consumer Electronics Magazine*, 10(4):63–68, 2021.
- [15] Bansilal Bairwa, Sagar Rathod, Udaykumar R Yaragatti, and KA Manohar. Development of fault detection method in cable using arduino uno. In *2022 IEEE 7th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, volume 7, pages 472–476. IEEE, 2022.
- [16] Yemeserach Mekonnen, Srikanth Namuduri, Lamar Burton, Arif Sarwat, and Shekhar Bhansali. Machine learning techniques in wireless sensor network based precision agriculture. *Journal of the Electrochemical Society*, 167(3):037522, 2019.
- [17] Bansilal Bairwa, K Yogeswar Reddy, D Deva, K Kavya, and KA Manohar. Development of iot based gsm controller for agriculture applications. In *2022 Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT)*, pages 1–6. IEEE, 2022.
- [18] Bansilal Bairwa, Sanjeeva Kumar, J Niharika, BS Sagar, R Nithyashree, S Prem Kumar, and CN Raghu. Modeling and simulation of solar irrigation system with im drive using pwm controller. In *AIP Conference Proceedings*, volume 2461, page 050001. AIP Publishing LLC, 2022.
- [19] Swapna Mansani, Santoshkumar Hampannavar, KN Patil, K Venkata Lakshmi Narayan, Bansilal Bairwa, and Endalew Ayenew. Wind speed assessment of three climate zones of ethiopia. In *2021 Innovations in Power and Advanced Computing Technologies (i-PACT)*, pages 1–4. IEEE, 2021.